

Supplementary information

A scRNA-seq atlas of chronic inflammatory skin diseases

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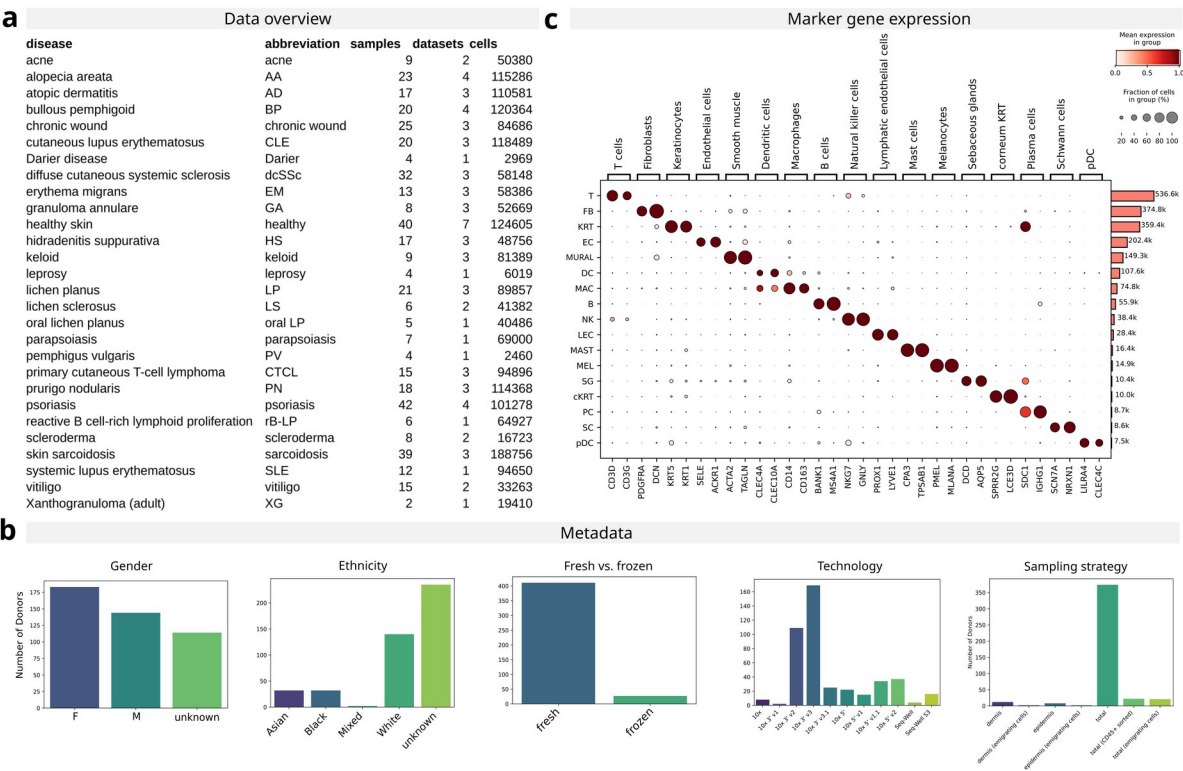
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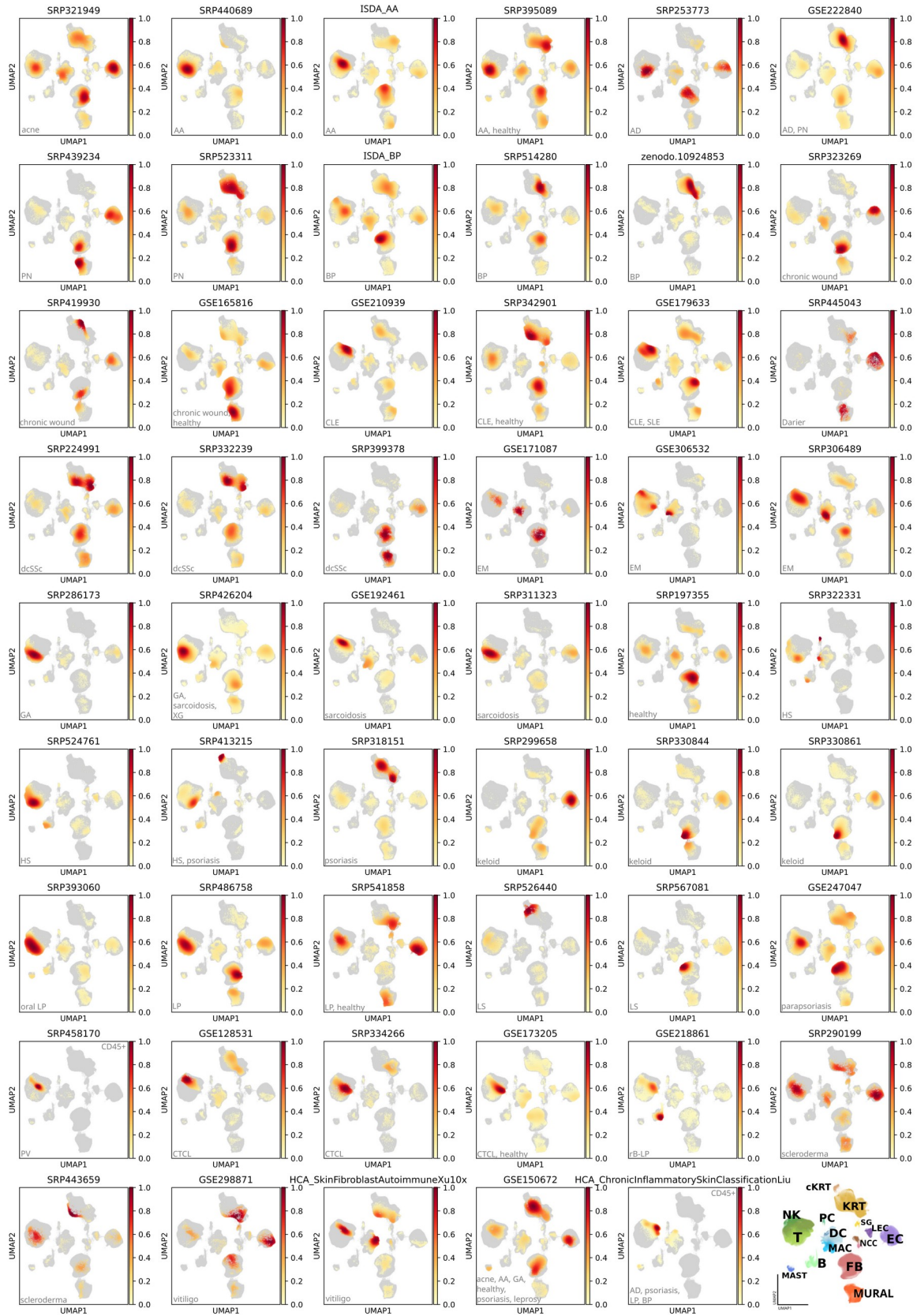
Key points:

- diseased skin atlas, scRNA-seq, cutaneous inflammation, inflammatory skin diseases

Abbreviations: AA, alopecia areata; AD, atopic dermatitis; BP, bullous pemphigoid; CLE, cutaneous lupus erythematosus; CTCL, primary cutaneous T-cell lymphoma; dcSSc, diffuse cutaneous systemic sclerosis; EM, erythema migrans; GA, granuloma anulare; HS, hidradenitis suppurativa; LP lichen planus; LS, lichen sclerosus; PN, prurigo nodularis; PV, pemphigus vulgaris; rB-LP, reactive B cell-rich lymphoid proliferation; SLE, systemic lupus erythematosus; XG, xanthogranuloma

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56 *Density distributions for each dataset are presented as individual plots. The associated disease for each dataset is indicated*
 57 *in the lower left corner of the respective plot. The final plot provides an overview of cell type annotations.*
 58 *B, B cells; DC, dendritic cells; EC, endothelial cells; FB, fibroblasts; KRT, keratinocytes; LEC, lymphatic endothelial cells;*

59 *MAC, macrophages; MAST, mast cells; MURAL, mural cells; NCC neural crest cells; NK, natural killer cells; PC plasma cells;*
60 *SG, sweat glands*



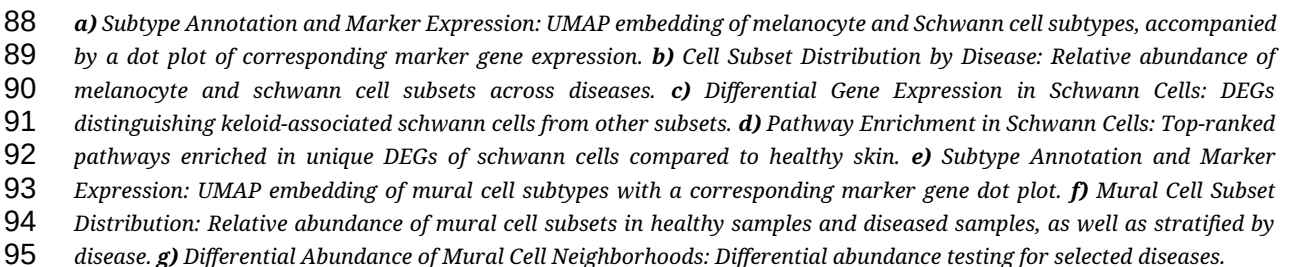
67 *testing for selected diseases. Each dot represents a neighborhood of cells, with significantly enriched neighborhoods (FDR <*
68 *10%) highlighted in black.*

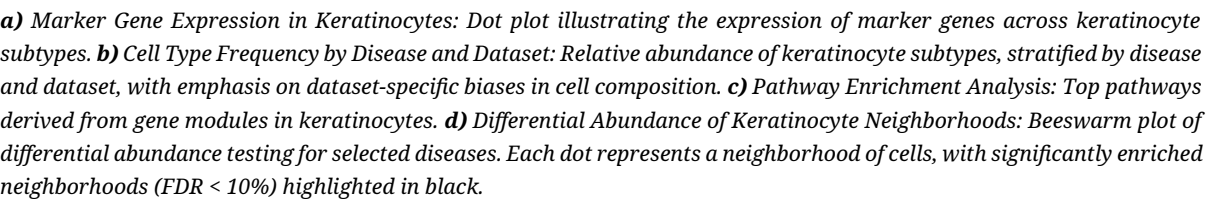
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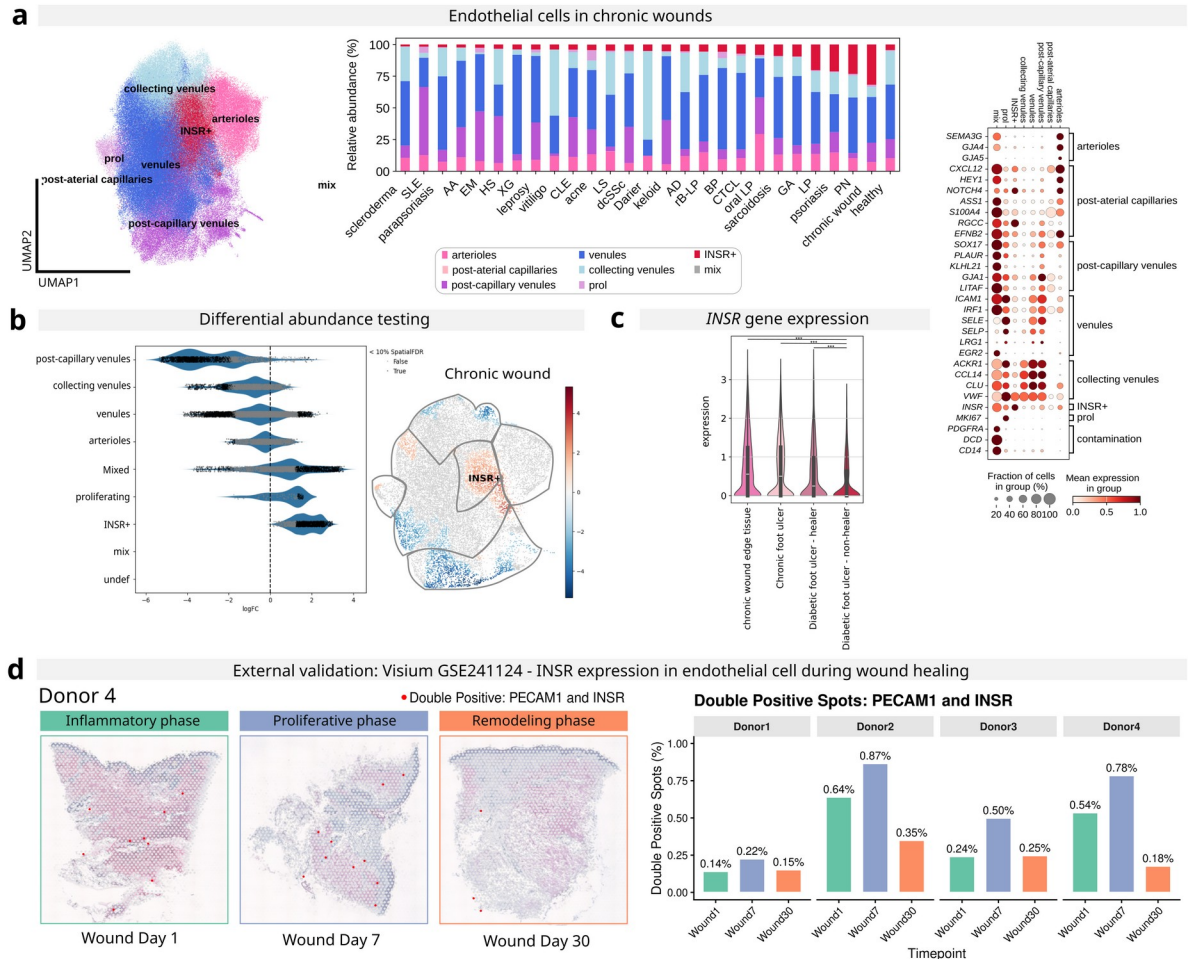






Differential Abundance of Myeloid Neighborhoods: Beeswarm and UMAP plots of differential abundance testing for selected diseases. Each dot in the beeswarm plot represents a neighborhood of cells, with significantly enriched neighborhoods (FDR < 10%) highlighted in black. **d)** Pathway Enrichment in Myeloid Cells: Top pathways derived from myeloid gene modules.

Supplementary Figure 9: Endothelial cells

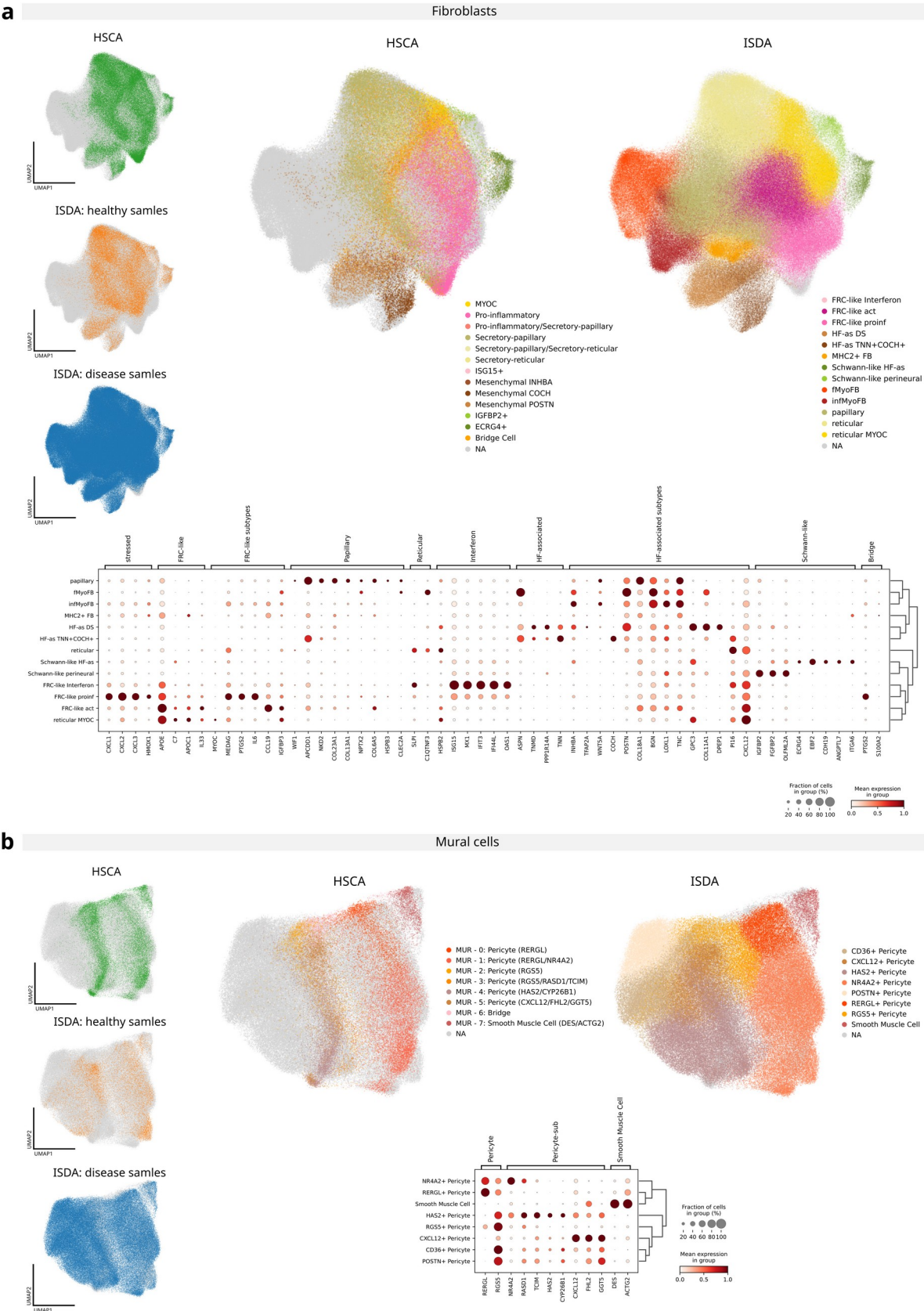


a) Endothelial Cell Subtype Characterization: UMAP embedding of endothelial cell subtypes, relative cell type distribution across diseases, and a dot plot of marker gene expression. **b) Differential Abundance in Chronic Wounds:** UMAP and beeswarm plot of differential abundance testing for chronic wounds. Each dot represents a neighborhood of cells, with significantly enriched neighborhoods (FDR < 10%) highlighted in black. **c) INSR Expression in Chronic Wound Subsets:** Expression levels of INSR across endothelial cell subsets in chronic wounds. **d) External Validation in Spatial Data (GSE241124):** Representative hematoxylin and eosin (H&E) stained sections depicting the distinct wound healing phases in donor 4. PECAM1/INSR double-positive spots are highlighted in green. The accompanying bar plot quantifies the relative frequency of double-positive spots across donors and time points.

Extended Data

Extended Data 1 - Harmonized cell annotation through the Healthy Skin Atlas

To achieve robust cell-type annotation of specialized subpopulations, we leveraged marker gene profiles from the Healthy Skin Cell Atlas (HSCA) and validated annotations via integration between HSCA and our Inflammatory Skin Disease Atlas (ISDA). Mapping examples for fibroblasts and mural cells are shown in Extended Data Fig. 1a,b. Highlighting in both cell types that disease specific subtypes can be found.



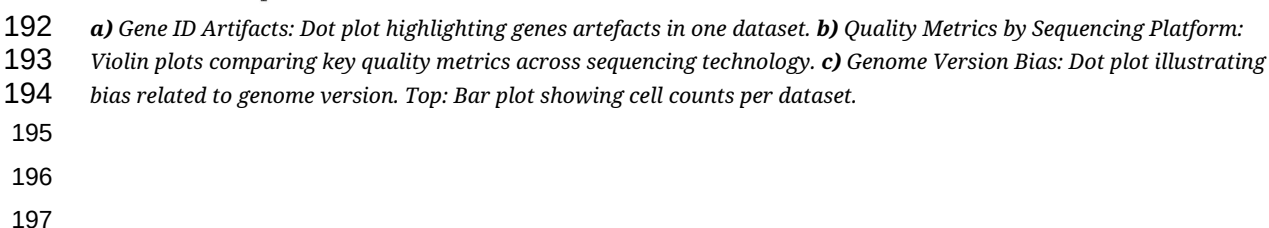
a,b) Integrated UMAP Embeddings of HSCA and ISDA Datasets.: Highlighting: HSCA cells (left top); healthy ISDA cells (left middle); disease ISDA cells.; cell annotation of HSCA cells (middle), cell annotation ISDA cells (right); HSCA marker gene expression on ISDA cell annotation (bottom). HSCA, healthy skin cell atlas; ISDA, inflammatory skin disease atlas

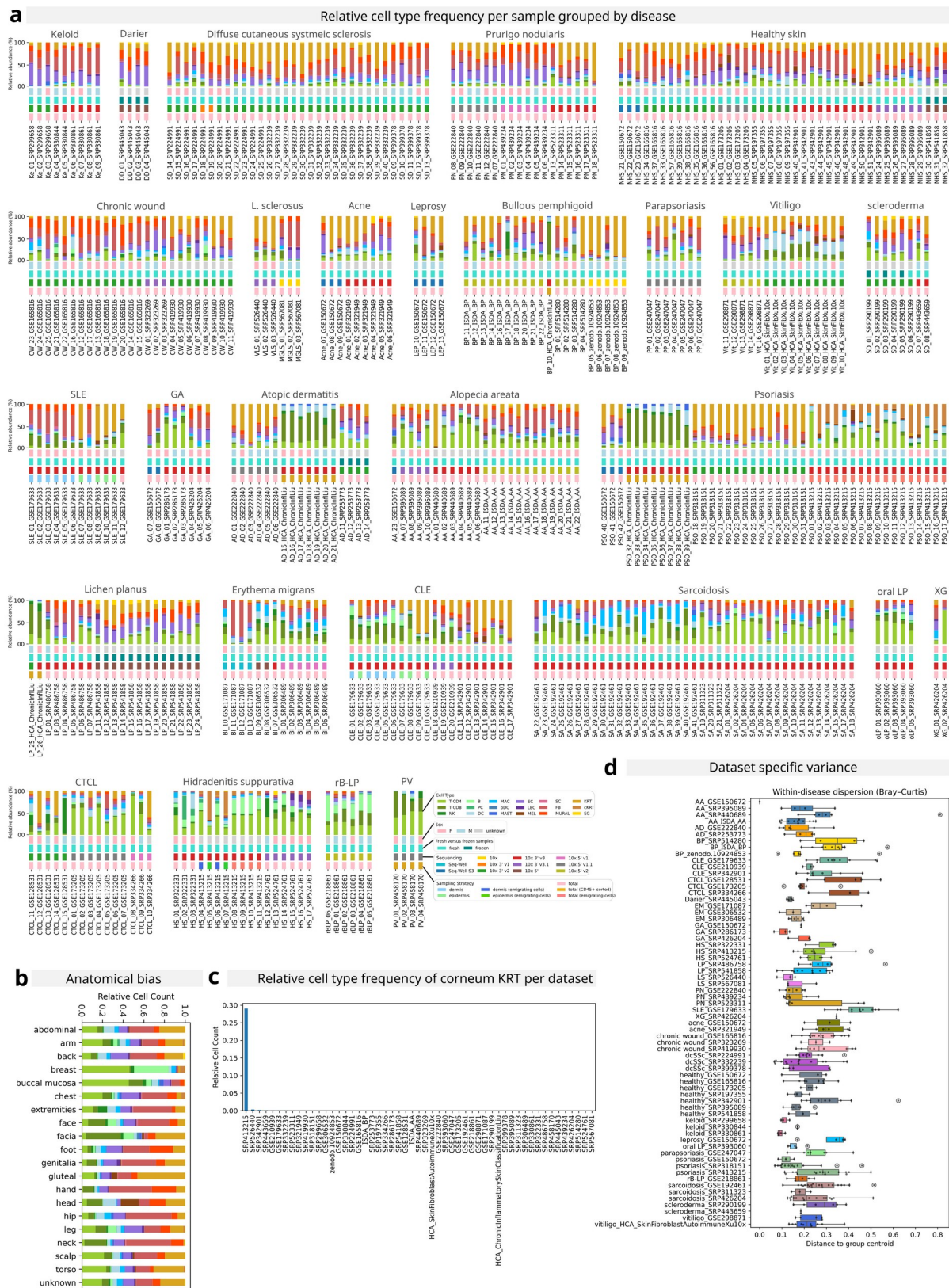
Extended Data 2 - Technical bias through sampling and bioinformatics methods

Sample preparation and data processing are critical steps in single-cell analyses, requiring careful consideration to minimize bias and ensure robust interpretation of results. Integrating this large amount of data enabled us to directly compare their influence on the results. During quality check of individual dataset we observed that a sample from GSE165816 had gene ID artifacts (dates embedded in identifiers; Extended Data Fig. 2a), a known issue in public datasets¹, underscores the critical need for rigorous preprocessing in meta-analyses. Apart from a higher mitochondrial content in Seqwell S3² samples, we did not observe major differences in choice of technology based on standard quality matrixes (Extended Fig. 2b). Two of the included datasets (GSE150672 and GSE17087) were aligned to the Human Reference Genome version 19 (Hg19), whereas all other datasets were aligned to Hg38 according to the respective methods. We deliberately included these datasets in order to assess the effect of different reference genomes. As expected, we observed that Hg19 aligned datasets used previous HGNC-approved nomenclature³ (Extended Fig. 2c). Surprisingly, this was also true for two further datasets (HCA_SkinFibroblastAutoimmuneXu10x and GSE128531) which were supposedly aligned to Hg38. This highlights the importance of consistent processing to prevent misinterpretations. Any findings in the downstream analysis representing the genome bias were excluded.

Sampling strategies introduced marked variability in cell type composition across diseases. For instance, lichen sclerosus samples showed diversity probably linked to sex and biopsy site (labia minora/interlabial sulcus vs. foreskin), while dataset variations in atopic dermatitis, lichen planus and scleroderma are likely linked to the usage of fresh vs frozen samples, which has a known impact on cell type recovery and especially keratinocytes⁴. Sarcoidosis samples from GSE192461 showed higher macrophage frequencies. Here, the authors first sorted for CD45+ and CD45- negative cells and then pooled them in an equal ratio⁵. Vitiligo samples from HCA_SkinFibAutoimmuneXu10x contained melanocytes, contrasting with disease hallmarks. Here, the authors took samples at the edge of lesions⁶ (Extended Fig. 3a). We did not observe major bias on cell type composition concerning bodysite location

(Extended Fig. 3b). Moreover, we observed a dataset-specific keratinocyte cluster that uniquely expressed corneum-associated genes like SPRR2G and LCE3D (Extended Fig. 3c, Supplementary Fig. 1c). These keratinocytes were found in two diseases (psoriasis and hidradenitis suppurativa), yet originated from the same dataset. Here, the authors analysed cells emigrating from the tissue after 48 hours of incubation⁷, which may have resulted in a processing-related artefact. This dataset was excluded from subsequent keratinocyte-related analyses. CLE and SLE samples showed high cell type variance (Extended Fig. 3d). Here, the authors split samples into dermis and epidermis⁸, reflected in different cell type compositions⁹. We also observed biological variation for diseases within datasets: xanthogranuloma, parapsoriasis, and leprosy displayed heterogeneity between samples and also for chronic wound, acne, bullous pemphigoid, alopecia areata, erythema migrans and CTCL heterogeneity seemed to be dataset independent (Extended Fig. 3a,d). Whereas oral lichen planus, rB-LP, pemphigus vulgaris and Darier disease samples appeared homogeneous (albeit single-dataset limited) and keloid and dcSSc appeared homogeneous even across datasets. Overall no major differences were observed based on sequencing technology but rather sampling strategy. These findings underscore the critical influence of sample preparation on cell type composition, a factor that must be considered when interpreting downstream results. Nevertheless, in all diseases cell type composition reflects known pathology, such as higher B cell abundance in HS.





a) Sample Metadata Overview: Stacked bar plots showing the relative frequency per sample, grouped by disease and stratified as follows (top to bottom): Cell type composition, Sex distribution, Fresh vs. frozen sample status, Sequencing technology, Sampling strategy. **b) Body site bias:** Stacked bar plots showing the relative frequency per anatomical region stratified by cell type. **c) Dataset-Specific Keratin Bias:** Highlighting the corneum KRT gene bias originating from a single dataset. **d) Cellular Composition Diversity:** Diversity index quantifying differences in cellular composition

between samples within each disease and dataset. Note: Samples containing CD45⁺-sorted cells were excluded from analyses in panel b,c).

Extended Data only references

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